

CS 340
Introduction to Machine Learning

Fall 2026

Instructor:	Prof. Ada Lovelace
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Office Hours:	Tue & Thu 2:00–3:30 PM, or by appointment
Meeting Time:	Mon, Wed, Fri 10:00–10:50 AM
Location:	Science Building, Lecture Hall B
Prerequisites:	CS 201 (Data Structures), MATH 250 (Linear Algebra), MATH 301 (Probability & Statistics)
Course Website:	https://cs340.university.edu

1 Course Description

This course provides a broad introduction to machine learning and statistical pattern recognition. Topics include supervised learning (linear regression, logistic regression, neural networks, support vector machines, decision trees, and ensemble methods), unsupervised learning (clustering, dimensionality reduction, and generative models), and the theoretical foundations that underpin modern machine learning practice. Students will gain hands-on experience by implementing algorithms in Python and applying them to real-world datasets.

2 Learning Objectives

Learning Objectives

1. Formulate supervised and unsupervised learning problems and select appropriate models for a given task.
2. Explain the bias–variance trade-off and apply regularization techniques to control model complexity.
3. Implement and evaluate core algorithms including linear regression, logistic regression, decision trees, and k -means clustering from scratch.
4. Apply neural network architectures using modern deep learning frameworks and diagnose common training issues (vanishing gradients, overfitting).
5. Perform rigorous model evaluation using cross-validation, ROC curves, and appropriate performance metrics.
6. Communicate experimental results clearly in written reports, including reproducible code and well-justified conclusions.

3 Required Textbooks and Materials

- Christopher M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006. (Primary reference)
- Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 3rd ed., O'Reilly, 2022.
- Additional readings and lecture slides will be posted on the course website.

Software: Python 3.11+, NumPy, scikit-learn, PyTorch (or TensorFlow), Jupyter Notebooks. All required software is available through the university computing labs or via free open-source installations.

4 Grading

Your final grade is determined by the weighted components shown below. All assignments are graded on a 0–100 scale; letter grades follow the standard university curve.

Component	Weight
Homework (4 assignments)	20%
Programming Projects (2)	25%
Midterm Examination	20%
Final Examination	25%
Participation & Peer Review	10%
Total	100%

5 Course Policies

5.1 Attendance

Attendance at lectures is expected but not formally recorded. You are responsible for all material covered in class, including announcements and schedule changes. Attendance at exams is mandatory; makeup exams are granted only for documented medical or family emergencies.

5.2 Late Work

Each student has a total of five **late days** that may be applied to any homework or project deadline, at most two per assignment. Once late days are exhausted, submissions are penalized 10% per calendar day late. No work is accepted more than seven days after the deadline.

5.3 Academic Integrity

All submitted work must represent your own effort unless explicitly authorized as a collaborative assignment. You may discuss ideas at a high level but must write all code and prose independently. Copying from any source—including online resources and other students—without proper attribution constitutes academic dishonesty and will be reported to the Office of Student Conduct. Violations result in a score of zero on the affected assignment for the first offense and a failing course grade for subsequent offenses.

5.4 Accommodations

Students requiring accommodations should register with the Disability Resource Center and contact the instructor within the first two weeks of the semester.

6 Course Schedule

Week	Dates	Topic
1	Sep 1–3	Course overview; linear algebra review; probability refresher
2	Sep 8–10	Linear regression; least squares; gradient descent
3	Sep 15–17	Logistic regression; maximum likelihood; regularization
4	Sep 22–24	Bias–variance trade-off; cross-validation; model selection
5	Sep 29–Oct 1	<i>Midterm 1 review</i> ; neural networks: perceptron & back-propagation
6	Oct 6–8	Deep learning: activation functions, optimization, CNNs
7	Oct 13–15	Support vector machines; kernel methods
8	Oct 20–22	Decision trees & random forests; ensemble methods
9	Oct 27–29	Unsupervised learning: k -means, hierarchical clustering
10	Nov 3–5	Dimensionality reduction: PCA, t-SNE
11	Nov 10–12	Generative models: GMMs, autoencoders, introduction to GANs
12	Nov 17–19	Reinforcement learning basics; course review; <i>Final exam review</i>
Midterm Examination: October 8 (in class)		
Final Examination: December 10, 9:00–11:00 AM		